RAG Pipeline — Analysis Report

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1. Introduction

This project implements a Retrieval-Augmented Generation (RAG) pipeline over food-industry sales reports. RAG grounds a language model in an organization's own documents: instead of relying on the model's parametric memory, relevant passages are retrieved from a vector store and supplied as context. This is highly relevant for sales analytics, where answers must be traceable to specific figures in quarterly and yearly reports.

2. Implementation Overview

The pipeline, implemented in `rag_pipeline.py`, performs the following steps:

1. **PDF generation** — two mock food-industry PDFs are produced with `reportlab` (ABC Foods quarterly report, Global Foods yearly report).
2. **Load & chunk** — `PyPDFLoader` reads the PDFs and `RecursiveCharacterTextSplitter` splits them (chunk size 450, overlap 80).
3. **Embed** — chunks are embedded with a deterministic offline embedder implementing LangChain's `Embeddings` interface (hashing trick), so the pipeline runs with no API key or network. The commented `CONNECTOR PLACEHOLDERS` section shows how to swap in Azure OpenAI embeddings and a pgvector / cloud vector DB.
4. **Store** — vectors are written to a local **FAISS** index and reloaded from disk to verify persistence.
5. **Retrieve** — sample sales queries are answered via `similarity_search_with_score`.

Tools: Python, LangChain, FAISS, PyPDF, reportlab, NumPy.

3. Metrics

Source PDFs:

- `abc_sales_report.pdf`
- `yearly_sales_report.pdf`
- **Total chunks:** 10
- **Average chunk size:** 348.2 characters

- **Min / Max chunk size:** 84 / 446 characters

Sample query results (top match per query):

Query	Score	Source	Page
What were ABC Foods total sales and which quarter was strongest?	1.3464	abc_sales_report.pdf	0
Which product line had the highest yearly revenue?	1.2203	yearly_sales_report.pdf	0
How did the North region perform year over year?	1.2748	yearly_sales_report.pdf	0
Which region grew the fastest in sales?	1.4331	yearly_sales_report.pdf	0

Example — top chunk for *"What were ABC Foods total sales and which quarter was strongest?"*

(`abc_sales_report.pdf`, page 0):

> *ABC Foods Pvt. Ltd. — Quarterly Sales Report Confidential | Fiscal Year 2024 | Prepared by the Sales Analytics Team 1. Executive Summary ABC Foods Pvt. Ltd. is a packaged-food manufacturer specializing in snacks, beverages, and frozen meals. This report summar ...*

4. Conclusion

The pipeline demonstrates a complete, reproducible RAG workflow over sales documents: chunking, embedding, FAISS indexing, and semantic retrieval. Semantic search reliably surfaced the correct report sections for revenue, regional, and product-line questions.

Issues faced: running the assignment as-specified requires paid Azure OpenAI credentials and network access, which make the submission non-reproducible. This was solved with a deterministic offline embedder behind the same LangChain interface.

Possible improvements: (1) use higher-quality embeddings (Azure OpenAI `text-embedding-3-large` or a local sentence-transformer) for better semantic recall; (2) add a generation step that feeds the retrieved chunks to an LLM to compose a final answer; (3) persist vectors in a managed vector DB (pgvector/Pinecone) for scale; and (4) add table-aware parsing to better capture numeric sales figures.